

tion (OCP) shuts down any circuit that pulls a power higher than the power limit. There are other in-built features like over-temperature protection (OTP), overload protection (OLP), short-circuit protection (SCP) and no-load operation (NLO) available in different models.

VI quadrants. There are three types of power supplies: single-quadrant supply that has output channels capable of only sourcing current; two-quadrant supply that can source and sink current in one voltage polarity; and four-quadrant power supply that is capable of sourcing and sinking currents in both positive and negative voltages.

Speed and accuracy. The speed with which voltage output can be varied and current be measured affects the total output capacity, especially for a large production system. Some power supplies are provided with test extensions that provide high-speed reading with good accuracy, especially for viewing high-speed transient or pulse events.

Buying a power sensor

As mentioned earlier, a high-end power meter generally has an external power sensor associated with it. Due to the large variance in product offerings and specifications, a detailed comparison is the best way to evaluate power sensors before making a purchase decision. Listed below are some of the factors to consider while buying a power sensor.

1. Basic factors like frequency range, power range, dynamic range, accuracy and speed of measurement required for the potential application(s)

2. Type of power to be measured —

Features of Power Measurement Devices

Device	Model	Brand	Features
AC power supply	AC6800	Keysight	- Intuitive user interface - Flexible I/O—LAN/LXI core and USB (standard) and GPIB (optional). Access and control the source remotely via a standard browser - Low cost of ownership
DC power supplies	N6900 & N7900	Keysight	VersaPower architecture: - Accelerates test-system throughput with industry-leading speed - Captures DUT's current profile with accurate measurements - Reduces ATE development time and cost with highly-integrated capabilities
DC power supply	N8900	Keysight	- Auto-ranging, single-output programmable DC power for ATE applications 14 different voltages, current and power combinations available - Easily parallel units to create 'one' power supply with >100kW of power
HDO power analyser	HD4096	Teledyne LeCroy	- Automatic set-up and display of relevant waveforms and parameters - Power device performance analysed in-circuit - Measurement and viewing of the time domain response of the entire control loop - Line analysis including line harmonics
Power analyser	PA4000	Tektronix	- Dual spiral shunt system for 0.05 per cent basic accuracy - Colour graphics display including waveforms and harmonics - Application based measurement modes and PC software
Power scope	PX8000	Yokogawa	- Power analyser + oscilloscope - Provides high-accuracy time based power measurement - Measures voltages up to 1000V RMS and currents up to 5A RMS (without external sensors), with a basic accuracy of ± 0.1 per cent
Power sensor	MA24105A	Anritsu	- Standalone, low-cost, plug-and-play device - Covers all major cellular and communication bands (WLL, GSM/EDGE, CDMA/EV-DO, WCDMA/HSDPA, WiMAX and TD-SCDMA) - Broad frequency range (350MHz to 4GHz)
Power sensor	U2040 X	Keysight	- Widest dynamic range found in power sensor (-70dBm to +26dBm) - Can make both average and time-selectivity average power measurements - Four USB models for wireless and radar applications, and a dedicated LAN model for satellite testing

continuous wave (CW), average or peak power

3. Pulse width, pulse repetition interval (PRI) and other pulse parameters requirement in case of pulse signals

4. Internal zero and calibration support

5. Voltage standing wave ratio (VSWR) specification

6. Triggering modes

7. PC-connectivity software and tools

Boosting efficiency and clean energy

We are seeing several initiatives across the globe with an aim to use power more efficiently—the '20-20-20' goal put forward by the European Union (EU) to reduce the present primary energy consumption by 20 per cent by the year 2020, and the transition of places like Djibouti, Tuvalu and many parts of Germany into 100 per cent energy-efficient regions, to name

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a few. This drive for a new, stable and affordable energy landscape, along with the changing policies, has opened up immense possibilities for the near future and has compelled many big players in the test and measurement industry to join the power sector. With such enormous trends happening in the industry, we can expect more innovations and breakthroughs in power measurement sector. ●

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